NCTU-EE IC LAB – Fall 2014

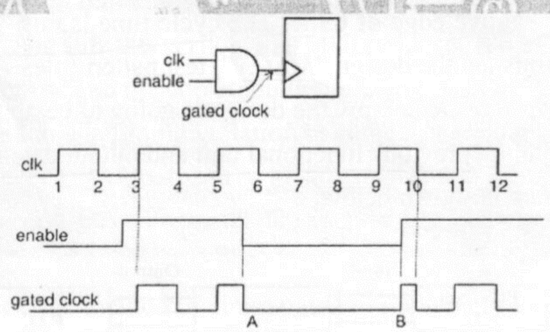
Online Test2: 13:30-16:20, 15th Jan at ED415

- ✧ It's an open book exam. If you add any assumption, please also state it clearly.
- ✧ Write the answers either in Chinese/English on the answer sheet.
- ✧ Hand in the **answer + question sheet** when finished.
- ✧ "※" in the beginning denotes the problem needs to read files of "OT2_FILE" in workstation. Please copy and untar "~iclabta01/OT2_FILE.tar".

Part 1

1. Clock gating is one of the most popular and effective method in low power design to reduce switching rate. Explain how it works. (5%)

   The following clock gating circuit may catch the wrong data if there exist a glitch in the enable signal. Please explain why and how to fix it? (5%)



2. Why does the clock source usually set as an **ideal clock** during synthesis in the design flow? (3%)

3. Sometimes, there might be a large arithmetic operator in the data path. Because of the delay through the path, the data cannot reach the destination within one cycle (resulting in violation of setup requirements). One option is to **reduce the clock frequency or do pipelining**, explain the drawback of this approach. (3%)

   Declaring the path as multicycle will allow the data more than one cycle, so that the data can reach the destination. However, just declaring the path as multi cycle is not sufficient. Write down what should we also do **in the RTL level**? (4%)

4. List one enhancement in **design** and **verification** using SystemVerilog, respectively. Moreover,

具体

briefly explain what the corresponding **advantage** is. (Assertion is excluded)(4%)

How can **assertions** help us in building verification environment? (4%)

5.

a). What is the purpose of the **"uniquify"** command in the design flow? (3%)

b). Please explain what's the **electro-migration and IR drop phenomena**, respectively. How to prevent these phenomena? (3%)

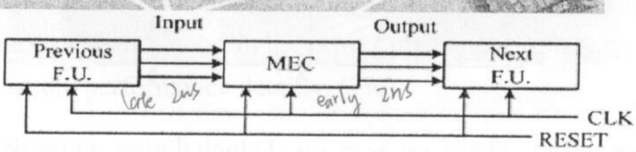
What's process antenna effect and how can APR tools fix antenna violation? (3%)

c) Why does the pad need power? (3%) What is the purpose of corner pad and io filler, respectively? (2%). How do we arrange the io pad and power pad around the core? (4%)

d) What is the purpose of placement blockage? How and where do we set the blockage? (4%)

Part2

6. (in 06_Zoro/) Zoro has implemented a design which can check whether a matrix is a parity matrix or not. (Where the parity property means that each row and each column has an even sum). This design is synchronous to the positive edge of CLK while the input data and output data are given and sampled at the negative edge of CLK. The cycle time is 6ns. The design has already been done by Zoro, who finished the design "*MEC.v*", test pattern files "*PATTERN.v*", "*TESTBED.v*" and the synthesis script "*syn.tcl*". Now the design is going to be applied in a project where input data are provided from the previous functional unit and output data are fed to the next functional unit. The block diagram is shown below:

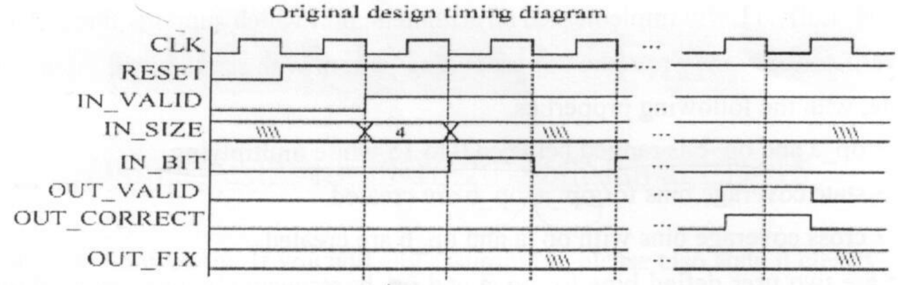


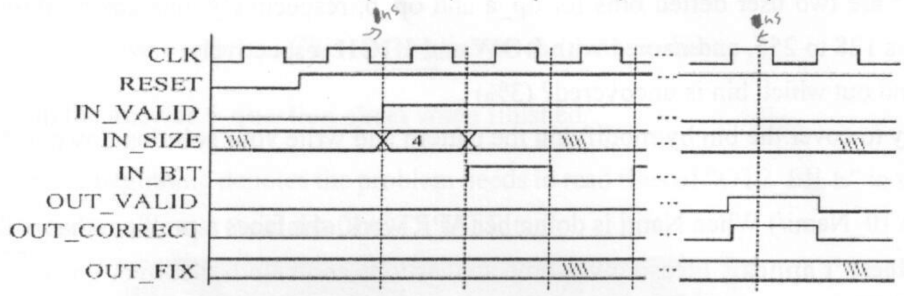
The clock cycle time is still **6ns**. However, these two external functional units can not satisfy the timing constraint to provide you input on time and sample your output on time. The input data from the previous functional unit is late for **2ns** and the next functional unit needs to obtain your output data earlier for **2ns**. The timing diagram is shown below:

dwol.sldb → dw_foundation.sldb

0 1 2 3

Original design timing diagram





New design timing diagram

a). Modify the timing constraint to synthesize and write down the constraint you have modified. (4%)

b). Modify the test pattern to make sure the gate-level netlist can satisfy this situation, and write down the statements you have modified. (4%)

7. ※(in 07_Usopp/) Usopp had implemented a multiplier. The function is just OUT = A * B. Usopp had implemented the design in the verilog file MUL.v. Also, his collegue Perona had written down the synthesis script file and the corresponding pattern. But you have some trouble now: The timing does not met and there is timing violation in gate level simulation. The project leader says that clock period should be fixed to 4ns. Try to solve these two problems and write down what you have modified using **multicycle approach.** (8%)

8. ※(in 08_Sanji/) Sanji use the method of clock-gated to implement a register file and also write a synthesis script and using PrimeTime-px to perform the power analysis.

a) Now try to compare the power of the design with gated clock and that without gated clock, and briefly describe your camparison flow (ex: you have modify about RTL code or synthesis script) and write down the power improvement in percentage.(4%)

b) Latch-based clock gating is a popular way to implement clock gating, usually the latch and the AND(OR) gate are further put together in a cell, called Integrated Clock Gating cell. Try to perform this clock gating method by modifying the synthesis script and write down what you have modify and the power improvement in percentage. (Hint: Use the synthesis command you have learn in the lab, and you will find the Integrated Clock Gating cell in the netlist) (4%)

9. ※(in 09_Luffy/) Luffy implemented an arithmetic unit which supports three kinds of operations. The verification team wrote the corresponding pattern with randomized inputs: op_a, op_b, and opcode, with the following properties.

- op_a and op_b is ranged between 0 to 15 while multiplying
- state coverage bins for op_a, op_b are created

coverage bins

- cross coverage bins with op_a and op_b are created.

There are two user defied bins for op_a and op_b, respectively, one covers 0 to 127, the other covers 128 to 255, and named with **LOW** and **HIGH** respectively.

a) Find out which bin is uncovered? (3%)

b) Try to cover the bin by modifying the pattern and write your solution down. (3%)

10. ※(In 10_Nami/) When Nami is doing her APR work, she faces a problem that is **the core power pin doesn't appear**. Please give some suggestions and help her correct this problem. (*You can restore CHIP_connect.enc*) (5%)

11. ※(In 11_Brook/) After days of hard work, Brook have completed an APR design with hard macro. Restore the design CHIP_OT2.enc.

a) There are some drawbacks of the floor planning and power planning. Point out 1 drawback of the floor plan. (3%)

b) Use another terminal to open IR drop result. Key in the following command under 11_Brook:

% firefox Power_log/PD_25C_dynamic_1/Reports/ CHIP.main.html &

Click VDD in the POWER NETS section and then 1.315V 1.639V 1.800V in the IR DROP ANALYSIS section, you will find the IR drop result of the deisgn.

According to the IR drop results, please give your comment on the IR drop distribution and some probable reasons (i.e. what's not good in the power planning). Finally, give some suggestions to the design to improve the performance detailly. (7%)

12. ※(In 12_Franky/) Restore the design CHIP_OT2.enc in 12_Franky. This design has just ran the postCTS timing analysis.

(1) Open timingReports/CHIP_postCTS.summary, you may find some problems of DRVs and timing slack.

(2) Verify geometry and connectivity you'll find DRC and LVS violations of the design.

Explain the reasons that cause the above violations and slacks in detailed, and give 2 solutions to fix the problem without changing the chip size. (5%)

(Hint: In encounter.log file, search the keyword "ERROR" you may findout the reason that cause the above violations and negative slacks.)